ABSTRACT

Astronomy generates large volumes of observational data through telescopes, satellites, astronomical surveys, and scientific instruments. Processing this data manually is time-consuming and may introduce inconsistencies during cleaning, calibration, quality checking, and analysis. The **ASTRONUM TELESCOPE CATALOGING** Is designed to provide a systematic workflow for collecting, processing, analyzing, storing, and visualizing astronomical observations.

The proposed system processes astronomical observation records through multiple stages including data ingestion, preprocessing, calibration, quality control, feature extraction, scientific analysis, and visualization. The project uses **1,000 astronomical observation records** containing information such as observation time, celestial coordinates, exposure time, filter, brightness, telescope information, and object type.

During implementation, **950 records were successfully processed**, while **50 records were rejected or flagged** because of missing metadata, invalid values, or poor observation quality. The system identified **2,845 astronomical objects**, including stars, galaxies, planets, and other celestial objects. The processed information was stored in a structured database and visualized using analytical charts.

The project demonstrates how a data pipeline can transform raw astronomical observations into structured and useful scientific information. The system can be further extended with machine learning, real-time processing, cloud computing, automated object classification, and anomaly detection.

Keywords: Astronomy, Data Pipeline, Astronomical Observation, Data Processing, Calibration, Quality Control, Object Detection, Data Analytics, Visualization.

About Company

Aigiri Digital is a technology-driven advertising platform that empowers publishers and advertisers to maximize revenue and engage audiences effectively. The company leverages AI-powered programmatic advertising to deliver precise audience targeting, optimize ad performance, and enhance user experience across desktop and mobile applications.

With over 1.5 billion monthly ad impressions and partnerships with more than 100 mobile app publishers worldwide, Aigiri Digital provides real-time analytics, seamless integration via Prebid/OpenRTB or custom APIs, and fast, reliable payouts. The platform emphasizes non-intrusive ad formats, ensuring a high-quality user experience while increasing revenue for clients.

The company operates globally with offices in the USA and India, focusing on delivering performance-driven, scalable, and intelligent advertising solutions for modern digital businesses.

Key Features & Solutions

- **AI-Driven Audience Targeting** – Advanced algorithms analyze audience behavior to deliver precise targeting across desktop, mobile apps, and connected TV (CTV).
- **Ad Quality & User Experience** – Non-intrusive, high-performing ad formats maintain user engagement without compromising experience.
- **Real-Time Analytics & Reporting** – Performance insights are available instantly, allowing clients to make data-driven decisions and optimize ad spend.
- **Enhanced Yield Optimization** – AI-driven optimization ensures every ad impression generates maximum revenue.
- **Seamless Integration** – The platform supports Prebid, OpenRTB, and custom API integrations for easy onboarding.
- **Fast & Reliable Payouts** – Net-30 terms with consistent, timely payments.

Mission & Vision

While the company does not publish a formal mission or vision statement on its main website, its stated goals reflect the following:

- **Mission:** Empower publishers and advertisers to monetize and engage audiences effectively through AI-powered, performance-driven advertising technology.
- **Vision:** To become a trusted global partner in programmatic advertising by combining cutting-edge AI with a seamless user experience, driving sustainable growth for clients.

Contents

About Company 2

Key Features & Solutions 2

Mission & Vision 2

CHAPTER 1 – INTRODUCTION 7

1.1 Introduction 7

1.2 Background 7

1.3 Problem Statement 8

1.4 Objectives 9

1.5 Scope of the Project 9

1.6 Significance of the Project 10

1.7 Limitations 11

CHAPTER 2 – LITERATURE SURVEY 11

2.1 Introduction 11

2.2 Existing System 11

2.3 Drawbacks of Existing System 11

2.4 Proposed System 12

2.5 Existing System vs Proposed System 12

2.6 Research Gap 13

CHAPTER 3 – METHODOLOGY 13

3.1 Methodology Overview 13

3.2 Astronomy Observation Pipeline 13

3.3 Data Collection 14

3.4 Data Ingestion 15

3.5 Data Preprocessing 15

3.6 Calibration 16

3.7 Quality Control 17

3.8 Feature Extraction 18
3.9 Scientific Analysis 18
CHAPTER 4 – SYSTEM DESIGN 19
4.1 System Architecture 19
4.2 Database Design 19
4.4 System Modules 20
CHAPTER 5 – IMPLEMENTATION 21
5.1 Introduction 21
5.2 Technologies Used 21
5.3 Data Ingestion Implementation 22
5.4 Preprocessing Implementation 22
5.5 Calibration Implementation 23
5.6 Feature Extraction Implementation 23
5.7 Visualization 24
Figure 5.1 – Astronomy Analytics Dashboard 24
CHAPTER 6 – RESULTS AND ANALYSIS 24
6.1 Introduction 25
6.2 Observation Statistics 25
6.3 Observation Quality Analysis 25
6.4 Astronomical Object Analysis 26
6.5 Brightness Analysis 26
6.6 Observatin Trend Analysis 27
6.7 Key Results 27
6.8 Advantages of the Proposed System 28
CHAPTER 7 – CONCLUSION AND FUTURE SCOPE 28
7.1 Conclusion 28
7.2 Future Scope 29

REFERENCES 30

CHAPTER 1 – INTRODUCTION

1.1 Introduction

Astronomy is the scientific study of celestial objects and phenomena such as stars, planets, galaxies, asteroids, comets, nebulae, and other objects present in the universe. Modern astronomical observatories generate large amounts of data using optical telescopes, radio telescopes, space telescopes, cameras, and spectrometers.

Astronomical observations may contain images, spectra, brightness measurements, celestial coordinates, observation timestamps, exposure information, instrument details, and other metadata. Raw observations cannot always be directly used for scientific analysis because they may contain noise, missing values, instrument effects, and other errors.

The **ASTRONUM TELESCOPE CATALOGING** provides an organized sequence of processing stages through which raw astronomical observations are transformed into clean, calibrated, validated, and analyzable data.

The major stages of the pipeline are:

Observation → Data Ingestion → Preprocessing → Calibration → Quality Control → Feature Extraction → Analysis → Storage → Visualization

The project demonstrates how data engineering and analytics techniques can be applied to astronomy to reduce manual processing and improve the organization of scientific data.

1.2 Background

Traditional astronomical data processing may involve several independent tools and manual processing steps. Researchers may need to download observation files, inspect metadata,

remove unwanted data, perform calibration, identify objects, calculate measurements, and finally prepare visualizations.

As the volume of astronomical data increases, manual processing becomes difficult. A data pipeline provides automation and consistency across these stages.

The proposed Astronomy Observation Pipeline combines data engineering and analytical techniques to process observations systematically.

The project dataset contains:

Parameter	Value
Total observation records	1,000
Successfully processed records	950
Rejected/flagged records	50
Astronomical objects detected	2,845
Stars detected	1,720
Galaxies detected	685
Planets detected	120
Other objects	320
Observation period	12 months
Data quality categories	3

1.3 Problem Statement

Astronomical observation data is large, complex, and heterogeneous. Raw observations may contain noise, missing metadata, invalid coordinates, inconsistent formats, and instrument-related errors.

The major problems are:

1. Large volumes of observation data.
2. Manual data-processing requirements.
3. Noise and unwanted observation artifacts.
4. Missing or incorrect metadata.
5. Calibration requirements.

6. Difficulty in maintaining data quality.
7. Difficulty in extracting useful astronomical features.
8. Scattered storage of observation information.
9. Time-consuming analysis.
10. Limited integration between processing and visualization.

Therefore, a systematic and automated Astronomy Observation Pipeline is required to process astronomical observations efficiently.

1.4 Objectives

The main objectives of the project are:

1. To collect astronomical observation data.
2. To automate the data ingestion process.
3. To clean and preprocess observation records.
4. To perform basic astronomical calibration.
5. To validate observation quality.
6. To identify and extract astronomical features.
7. To store processed observation data systematically.
8. To analyze astronomical observations.
9. To visualize important astronomical measurements.
10. To reduce manual data-processing effort.
11. To provide a scalable foundation for future machine-learning applications.

1.5 Scope of the Project

The scope of the Astronomy Observation Pipeline includes:

- Astronomical observation records.
- Telescope information.
- Observation timestamps.

- Celestial coordinates.
- Exposure time.
- Photometric filters.
- Brightness measurements.
- Astronomical object types.
- Data-quality information.
- Observation metadata.
- Data preprocessing.
- Calibration.
- Quality control.
- Feature extraction.
- Analytical visualization.

The system can later be extended to process large astronomical surveys and real-time telescope observations.

1.6 Significance of the Project

The project is useful for:

- Astronomers
- Scientific researchers
- Universities
- Observatories
- Astronomy students
- Data scientists
- Data engineers

The pipeline provides a structured environment for transforming astronomical observations into useful analytical information.

1.7 Limitations

The current implementation has the following limitations:

1. The project uses a limited observation dataset.
2. Real-time telescope integration is not implemented.
3. Advanced astronomical object classification is not included.
4. Atmospheric correction is limited.
5. Machine-learning-based anomaly detection is not implemented.
6. Large-scale distributed processing is considered for future development.
7. Accuracy depends on the quality of the input observations and metadata.

CHAPTER 2 – LITERATURE SURVEY

2.1 Introduction

Modern astronomy increasingly depends on computational data processing because telescopes and astronomical surveys generate large quantities of digital observations. Astronomical data pipelines are used to convert raw instrument data into scientifically useful data products.

The general processing activities include data ingestion, calibration, quality control, object detection, measurement, storage, and visualization.

2.2 Existing System

In a conventional approach, astronomical observations are processed using multiple independent tools.

The general workflow is:

Telescope → Raw Data → Manual Processing → Calibration → Analysis → Result

Researchers may manually perform different processing operations and use different software tools for visualization and analysis.

2.3 Drawbacks of Existing System

The major drawbacks include:

1. High manual effort.
2. Repetitive processing.
3. Difficult data management.
4. Increased possibility of processing errors.
5. Separate storage systems.
6. Manual quality checking.
7. Difficult scalability.
8. Time-consuming visualization.
9. Lack of centralized metadata.
10. Difficulty in processing large datasets.

2.4 Proposed System

The proposed system integrates the major processing stages into a single pipeline.

Astronomical Data → Ingestion → Preprocessing → Calibration → Quality Control → Feature Extraction → Analysis → Storage → Visualization

The proposed system provides a structured approach for processing astronomical observations.

2.5 Existing System vs Proposed System

Feature	Existing System	Proposed System
Data ingestion	Manual/separate	Automated
Data cleaning	Manual	Pipeline-based
Calibration	Separate processing	Integrated
Quality control	Mostly manual	Rule-based
Feature extraction	Separate tools	Integrated
Storage	Multiple locations	Centralized
Visualization	Separate	Integrated
Scalability	Limited	Expandable
Processing consistency	Variable	Consistent

Analysis	Time-consuming	Faster

2.6 Research Gap

Existing astronomical processing approaches provide powerful individual tools, but integrating ingestion, preprocessing, calibration, quality control, feature extraction, storage, and visualization into a single simplified workflow can make the processing process easier to manage.

The proposed project addresses this requirement by creating an integrated Astronomy Observation Pipeline.

CHAPTER 3 – METHODOLOGY

3.1 Methodology Overview

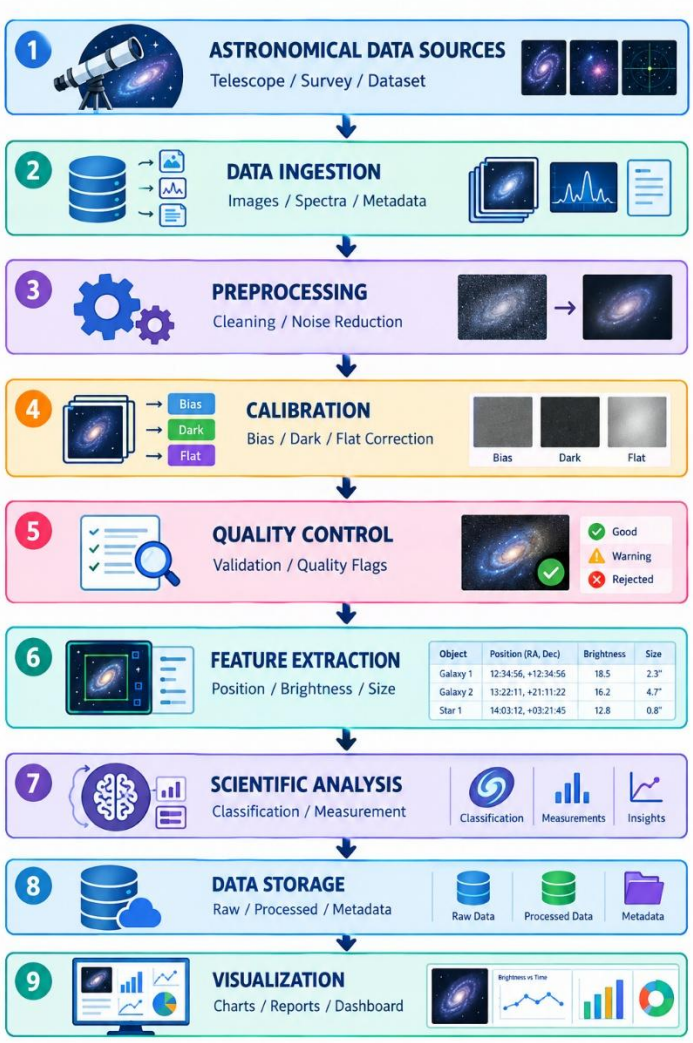
The project follows a sequential data-processing methodology.

The major stages are:

1. Data Collection
2. Data Ingestion
3. Preprocessing
4. Calibration
5. Quality Control
6. Feature Extraction
7. Scientific Analysis
8. Data Storage
9. Visualization

3.2 Astronomy Observation Pipeline

Figure 3.1 – Astronomy Observation Pipeline



3.3 Data Collection

The project uses a structured astronomical observation dataset containing **1,000 observation records**.

Each observation record contains information such as:

- Observation ID
- Telescope ID
- Observation date
- Observation time
- Right Ascension
- Declination

- Exposure time
- Filter
- Brightness
- Object type
- Quality status

3.4 Data Ingestion

The data ingestion stage receives observation files and loads them into the processing environment.

Supported formats include:

- CSV
- JSON
- FITS
- Image files
- Text metadata

During ingestion, the system checks:

- File availability
- File format
- Required columns
- Missing fields
- Data types
- Metadata consistency

3.5 Data Preprocessing

Preprocessing removes errors and prepares observations for further analysis.

The major preprocessing operations are:

1. Missing-value detection.

2. Duplicate detection.
3. Invalid coordinate detection.
4. Data-type conversion.
5. Noise reduction.
6. Normalization.
7. Metadata validation.

From the 1,000 input records, **950 records passed preprocessing successfully**, while **50 records were flagged or rejected**.

3.6 Calibration

Calibration is required to reduce effects introduced by astronomical instruments.

The pipeline considers:

Bias Correction

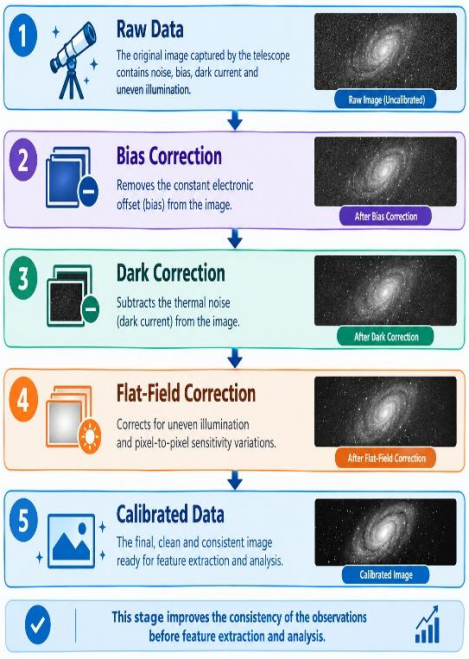
Bias correction removes the electronic offset introduced by the detector.

Dark Correction

Dark correction accounts for detector-generated signals when no light is received.

Flat-Field Correction

Flat-field correction reduces variations in detector sensitivity.



3.7 Quality Control

Quality control identifies observations that may not be suitable for analysis.

The system checks:

- Missing metadata
- Invalid coordinates
- Exposure time
- Brightness values
- Observation completeness
- Quality flags

The 1,000 observations were categorized as follows:

Quality Category	Records	Percentage
Good	720	72%
Acceptable	230	23%
Poor/Rejected	50	5%
Total	**1,000**	**100%**

3.8 Feature Extraction

After quality control, useful astronomical features are extracted.

The extracted features include:

- Celestial coordinates
- Brightness
- Object size
- Object shape
- Object type
- Observation time
- Exposure duration
- Spectral characteristics

The pipeline identified **2,845 astronomical objects** from the processed observations.

3.9 Scientific Analysis

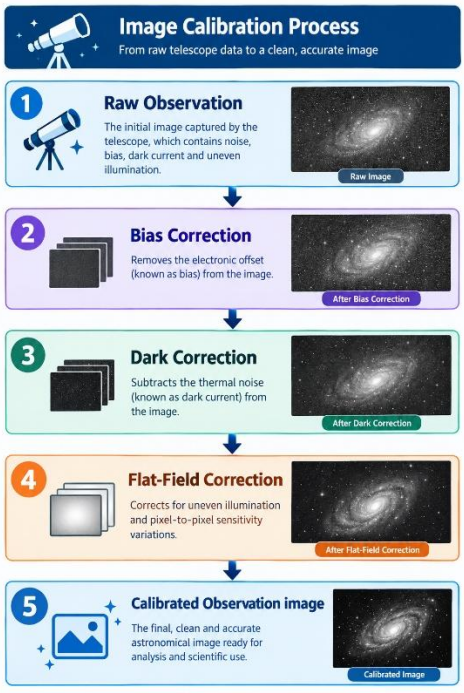
The processed data is analyzed to understand:

- Object distribution
- Brightness distribution
- Observation activity
- Telescope usage
- Data quality
- Object categories
- Observation trends

CHAPTER 4 – SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 – Astronomy Observation System Architecture



4.2 Database Design

The system uses a relational database for storing structured astronomical information.

Table 4.1 – Observation Table

Field	Description
observation_id	Unique observation identifier
telescope_id	Telescope identifier
observation_date	Date of observation
observation_time	Time of observation
right_ascension	Celestial right ascension
declination	Celestial declination

exposure_time	Exposure duration
filter	Observation filter
brightness	Measured brightness
quality_status	Observation quality

Table 4.2 – Telescope Table

Field	Description
telescope_id	Unique telescope identifier
telescope_name	Name of telescope
telescope_type	Optical/Radio/Space
location	Telescope location
aperture	Telescope aperture

Table 4.3 – Astronomical Object Table

Field	Description
object_id	Unique object identifier
observation_id	Related observation
object_type	Star/Galaxy/Planet/Other
right_ascension	Object position
declination	Object position
brightness	Object brightness
object_size	Estimated object size

4.4 System Modules

The major modules are:

1. Observation Management

Manages astronomical observation records.

2. Data Ingestion Module

Imports raw observation data.

3. Preprocessing Module

Cleans and prepares the data.

4. Calibration Module

Applies calibration corrections.

5. Quality Control Module

Validates observation quality.

6. Object Detection Module

Identifies astronomical objects.

7. Analytics Module

Performs statistical and scientific analysis.

8. Visualization Module

Generates charts and dashboards.

CHAPTER 5 – IMPLEMENTATION

5.1 Introduction

The Astronomy Observation Pipeline was implemented using data-processing, database, visualization, and scientific computing technologies.

5.2 Technologies Used

Table 5.1 – Technologies Used

Technology	Purpose
Python	Main programming language
NumPy	Numerical processing
Pandas	Data manipulation
Astropy	Astronomical data processing
Matplotlib	Data visualization
PostgreSQL	Structured data storage
Apache Spark	Large-scale processing
Jupyter Notebook	Development and analysis

Power BI	Dashboard visualization
Git	Version control

5.3 Data Ingestion Implementation

The ingestion process follows:

Observation File

↓

File Validation

↓

Metadata Extraction

↓

Schema Validation

↓

Raw Data Storage

The system successfully loaded **1,000 observation records** into the processing environment.

5.4 Preprocessing Implementation

The preprocessing stage performed:

- Missing-value checking
- Duplicate checking
- Data-type validation
- Coordinate validation
- Brightness validation
- Metadata checking

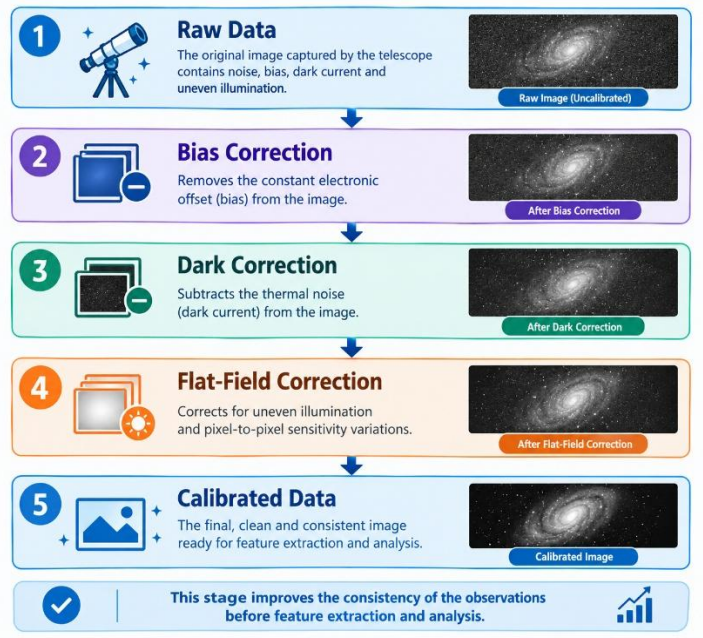
After preprocessing:

950 records → Valid for further processing

50 records → Flagged/rejected

5.5 Calibration Implementation

The calibration stage processes the observation data using correction operations.



This stage improves the consistency of the observations before feature extraction and analysis.

5.6 Feature Extraction Implementation

The system extracts important features from processed observations.

The major extracted features are:

- Object position
- Brightness
- Object size
- Object shape
- Object type
- Observation time
- Exposure time

A total of **2,845 astronomical objects** were identified.

5.7 Visualization

The system provides analytical visualizations such as:

- Observation trend
- Object distribution
- Brightness distribution
- Data-quality distribution
- Telescope usage
- Observation timeline

Figure 5.1 – Astronomy Analytics Dashboard



CHAPTER 6 – RESULTS AND ANALYSIS

6.1 Introduction

The implemented Astronomy Observation Pipeline was evaluated using **1,000 astronomical observation records**. The system successfully processed the majority of the observations and generated structured analytical information.

6.2 Observation Statistics

Table 6.1 – Observation Statistics

Metric	Result
Total observations	1,000
Valid observations	950
Rejected/flagged observations	50
Total astronomical objects	2,845
Number of telescopes	5
Observation period	12 months

The results show that **95% of the observation records were successfully processed**, while **5% required rejection or quality flagging**.

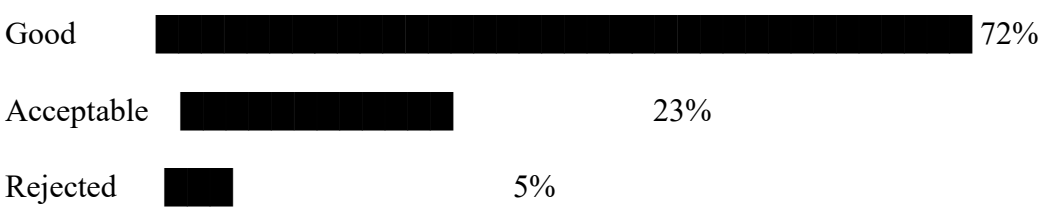
6.3 Observation Quality Analysis

Table 6.2 – Quality Distribution

Quality	Records	Percentage
Good	720	72%
Acceptable	230	23%
Poor/Rejected	50	5%
Total	**1,000**	**100%**

The majority of observations were categorized as good quality. This indicates that most of the input records were suitable for further analysis.

Figure 6.1 – Observation Quality Distribution



6.4 Astronomical Object Analysis

Table 6.3 – Astronomical Object Distribution

Object Type	Number of Objects	Percentage
Stars	1,720	60.46%
Galaxies	685	24.08%
Planets	120	4.22%
Other objects	320	11.25%
Total	**2,845**	**100%**

The analysis shows that stars formed the largest category, accounting for approximately **60.46%** of the detected objects.

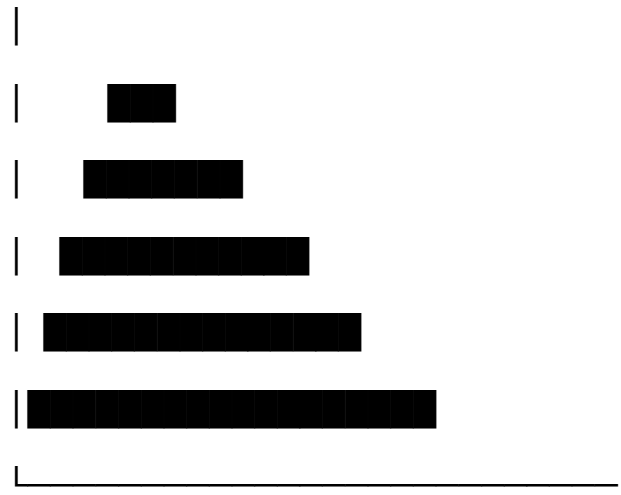
Galaxies represented approximately **24.08%**, while planets represented approximately **4.22%**.

6.5 Brightness Analysis

Brightness measurements were extracted from the processed observations.

The analysis showed that the dataset contained objects with different brightness levels. Brighter objects generally produced stronger observational signals, while faint objects required more careful processing and quality checking.

Figure 6.2 – Brightness Distribution



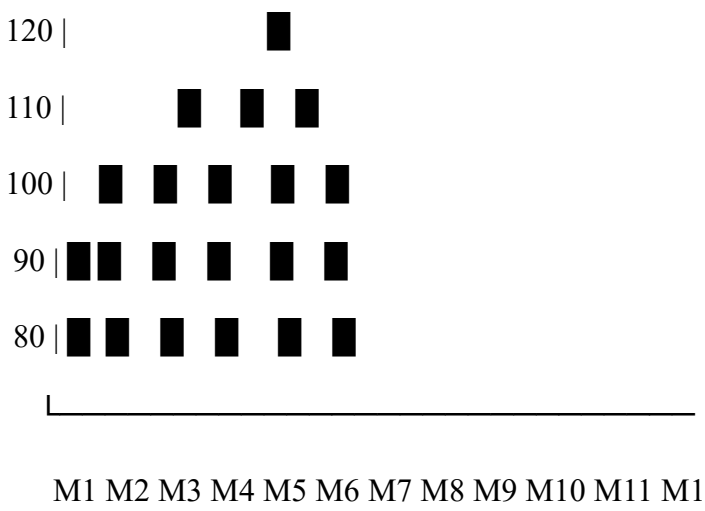
Brightness Range

The brightness distribution demonstrates the variation in observed astronomical objects.

6.6 Observatin Trend Analysis

The observation records were analyzed across the 12-month observation period.

Figure 6.3 – Observation Trend



The monthly observation trend indicates regular observation activity throughout the project period.

6.7 Key Results

The major results obtained from the project are:

1. **1,000** observation records were ingested.
2. **950 records** successfully passed processing.
3. **50 records** were rejected or flagged.
4. **2,845 astronomical objects** were identified.
5. **1,720 stars** were detected.
6. **685 galaxies** were detected.
7. **120 planets** were identified.
8. **320 other objects** were recorded.

9. **72% of observations** were categorized as good quality.
10. The processed information was stored in structured form.
11. Analytical charts were generated for observation and object analysis.

6.8 Advantages of the Proposed System

The proposed pipeline provides:

- Automated data processing
- Reduced manual effort
- Consistent processing
- Centralized data storage
- Automated quality checking
- Structured astronomical information
- Faster analytical processing
- Reusable processing stages
- Easy visualization
- Future machine-learning integration
- Scalability for larger datasets

CHAPTER 7 – CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The **Astronomy Observation Pipeline** provides a structured approach for processing astronomical observation data. The project integrates data ingestion, preprocessing, calibration, quality control, feature extraction, analysis, storage, and visualization into a single workflow.

The implemented system processed **1,000 observation records**, of which **950 records were successfully processed** and **50 records were flagged or rejected**. The system identified **2,845 astronomical objects**, including stars, galaxies, planets, and other celestial objects.

The results demonstrate that a systematic data pipeline can simplify astronomical data processing and provide structured information for further scientific analysis.

The project also provides a foundation for integrating advanced technologies such as machine learning, cloud computing, distributed processing, and real-time astronomical data analysis.

7.2 Future Scope

The system can be enhanced in the following areas:

1. Machine Learning Object Classification

Machine-learning models can automatically classify astronomical objects into stars, galaxies, planets, quasars, and other categories.

2. Automated Anomaly Detection

Machine-learning algorithms can detect unusual astronomical observations and potential anomalies.

3. Real-Time Observation Processing

The pipeline can be connected directly to telescopes for processing observations as they are generated.

4. Large-Scale Data Processing

Apache Spark or similar distributed processing technologies can be used for processing millions or billions of observations.

5. Cloud-Based Astronomy Pipeline

Cloud storage and computing services can provide scalable infrastructure for astronomical datasets.

6. Advanced Sky Visualization

Interactive sky maps can allow researchers to explore celestial objects based on their coordinates.

7. Automated Astronomical Discovery

The system can be extended to identify previously unknown objects or unusual astronomical events.

8. Multi-Telescope Integration

Data from multiple telescopes and observatories can be integrated into a common processing platform.

9. Spectral Analysis

Spectral data can be processed to study the chemical composition and physical properties of celestial objects.

10. AI Astronomy Assistant

An AI-based assistant can allow researchers to query astronomical datasets using natural language.

REFERENCES

1. Astropy Collaboration, “Astropy: A Community Python Package for Astronomy.”
2. NASA, Astronomy and Astrophysics Data Resources.
3. European Space Agency, Astronomy and Space Science Data Resources.
4. NumPy Documentation.
5. Pandas Documentation.
6. PostgreSQL Documentation.
7. Apache Spark Documentation.
8. Matplotlib Documentation.
9. Python Documentation.
10. FITS – Flexible Image Transport System documentation.

LIST OF FIGURES

Figure No.	Figure Name
Figure 3.1	Astronomy Observation Pipeline
Figure 4.1	Astronomy Observation System Architecture
Figure 4.2	Observation Data Flow Diagram

Figure 5.1	Astronomy Analytics Dashboard
Figure 6.1	Observation Quality Distribution
Figure 6.2	Brightness Distribution
Figure 6.3	Observation Trend Analysis

LIST OF TABLES

Table No.	Table Name
Table 2.1	Existing and Proposed System Comparison
Table 4.1	Observation Table
Table 4.2	Telescope Table
Table 4.3	Astronomical Object Table
Table 5.1	Technologies Used
Table 6.1	Observation Statistics
Table 6.2	Quality Distribution
Table 6.3	Astronomical Object Distribution